

Juan Martin Varela

juanmartinvarela2006@gmail.com | (786) 869-0940 | Miami, FL | Bright-Computing.com

EDUCATION

Duke University

May, 2029

B.S., Electrical & Computer Engineering and Computer Science

Durham, NC

AWARDS

- Amazon Future Engineer Scholar *Amazon, 2025*
- Coral Gables Community Foundation Scholarship *City of Coral Gables, 2025*

EXPERIENCE

Amazon SDE Intern

Jun. 2026 – Aug. 2026

Software Engineering Intern

Seattle, WA

Duke AERO

Aug. 2025 – Feb. 2026

Electrical & Software Engineering

Duke University

- Developed embedded firmware in **C++** for a liquid propulsion valve control system managing **8 servo actuators**.
- Implemented real-time data acquisition and processing from **12 onboard sensors using Python**, ensuring accurate telemetry and control feedback.
- Contributed to development of a 10km+ long-range live video transmission system. Built ground station GUI using **React Native**, enabling real-time visualization of telemetry and video data.

Coral Gables IT Department

Jun. 2024 – Aug. 2024

Software Engineering Intern

Miami, FL

- Collaborated in a team environment to build a **city weather web application** using the **ArcGIS API**.
- Performed large-scale data cleaning and transformation using **Python and JavaScript** to ensure reliable forecasting data.
- Engineered a **vector database for semantic search** of city archives using **Python and PyTorch**, improving document retrieval efficiency.

Tech Flex Leader - America On Tech, Inc.

Sep. 2024 – Apr. 2025

Member

Miami, FL

- Selected for a competitive year-long technical development program.
- Completed advanced coursework in full-stack web development (**HTML, CSS, JavaScript**).
- Trained in **data science** and machine learning using **Python and TensorFlow**.

ACADEMIC PROJECTS

- **EGR101**: Worked in a team to develop a distance measurement device for wheelchair users. Responsible for developing embedded firmware in **C++** for data acquisition and WiFi transmission.